Learning That Remembers You

An Open-Source AI-Native Learning Platform and Plugin Architecture for Longitudinal Learner Modelling at Scale

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[repository link redacted for review]
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Abstract

Traditional learning management systems (LMSs) deliver static, one-size-fits-all content with no longitudinal learner model and no adaptive tutoring. Intelligent tutoring systems (ITS) that do adapt remain either narrow-domain research prototypes or products disconnected from the course-hosting infrastructure most organisations already use. We present **Sudar**, a fully open-source, AI-native learning system released under the Apache 2.0 licence, making three contributions: (1) the **Sudar reference platform (LAMP)**, a working implementation unifying authoring, delivery, and intelligence around a persistent Digital Learner Twin, adaptive sequencing, six multimodal delivery formats, an AI tutor with longitudinal cross-session memory, bounded agent orchestration (SudarAgents), and consent-gated generative personalisation; (2) the **Adaptive Learning Layer (ALP)**, a novel plugin architecture enabling these capabilities to be deployed as independently installable services on top of existing LMSs without requiring platform replacement; and (3) an **economic analysis** demonstrating the full capability set can be delivered at a per-learner AI infrastructure cost below $0.02 per month, exceeding a 99% reduction relative to both incumbent commercial LMS licensing fees and proprietary AI provider stacks. The reference implementation is open source (Apache 2.0), grounded in a broad learning-science evidence base, and designed as an extensible bedrock on which the community can build additional modalities, intelligence plugins, and LMS connectors. A live deployment is accessible at *[demo link redacted for review]*.

Keywords: adaptive learning · intelligent tutoring · learner modelling · digital learner twin · open-source LMS · multimodal learning · LLMs · plugin architecture · AI-native education

1. Introduction

Learning management systems used by most organisations and universities deliver one-size-fits-all content: the same course for every learner, no memory of prior interactions, and no adaptation of sequence, difficulty, or support based on behaviour or prior knowledge. Research consistently shows that adaptive instruction and intelligent tutoring outperform fixed instruction, a gap quantified by Bloom's *two-sigma problem* [6], which demonstrated that one-to-one tutoring produces learning gains two standard deviations above classroom instruction. Yet mainstream LMS products do not maintain a longitudinal learner model or provide personalised, memory-aware tutoring [29, 1].

The inference cost collapse of the past four years changes the calculus entirely. AI inference costs have fallen approximately 1,000-fold from 2022 to 2026 [4], making it economically feasible, for the first time, to deliver a full personalised tutoring session to every learner for less than the cost of a single SMS message. Yet commercial LMS vendors continue to charge $5–$20 per user per month, with AI features offered as credits-based add-ons requiring enterprise contracts. This pricing structure excludes community colleges, NGOs, corporate L&D teams, and educational organisations in the Global South.

We address both the architectural and economic gap with **Sudar**, an AI-native learning system that: (1) implements a complete open-source reference platform with learner memory, adaptive paths, and a fully realised multimodal delivery stack; (2) is designed so its core components can be deployed as a layer on top of existing LMSs, the Adaptive Learning Layer (ALP); and (3) demonstrates that this capability set can be delivered at infrastructure costs approaching zero using open-weight models and zero-cost TTS.

Contributions:

- **LAMP (reference platform):** A fully open-source, working implementation of an AI-native learning system unifying authoring, delivery, and intelligence around a persistent Digital Learner Twin, adaptive sequencing, six multimodal delivery formats, bounded agent orchestration, and an AI tutor with longitudinal cross-session memory.
- **ALP (Adaptive Learning Layer):** A novel plugin architecture for deploying learner memory, adaptive tutoring, and modality choice as independently deployable services on top of existing LMSs, without infrastructure replacement.
- **Economic analysis:** Empirically observed infrastructure cost data demonstrating a 99%+ reduction relative to both incumbent commercial LMSs and proprietary AI provider stacks.

2. Background and Related Work

2.1 Foundational Learner Modelling

Bayesian Knowledge Tracing (BKT), introduced by Corbett and Anderson [9], models a student's latent mastery of individual knowledge components as a hidden Markov process, updating estimates from observed performance. The Digital Learner Twin in Sudar departs from BKT deliberately: rather than modelling discrete knowledge components for narrow curricula, it maintains a profile-level representation of learning style, modality affinity, behavioural engagement signals, and longitudinal tutor context, a practical, LLM-era interpretation of an open learner model [7] with longitudinal persistence, optimised for general-purpose course delivery across domains.

2.2 Adaptive Instruction and ITS

Adaptive learning systems that tailor content and difficulty to the learner improve outcomes compared to one-size-fits-all instruction [28, 3]. Persistent learner models allow systems to adapt over time [25, 7]. Bloom's two-sigma problem [6] provides the foundational motivation: if individual tutoring produces gains two standard deviations above classroom instruction, making high-quality adaptive tutoring universally accessible is one of the highest-leverage interventions available in education.

2.3 Multimodal Learning and Dual Coding

Dual coding theory and multimedia learning principles establish that presenting information across complementary modalities, verbal and visual, supports deeper encoding and accommodates individual differences [18, 8]. Content in Sudar is authored once and delivered in six modalities, with a modality dispatcher that recommends formats per learner profile, operationalising dual-coding principles at scale.

2.4 Conversational and LLM-Based Tutors

Khanmigo (Khan Academy) and Socratic (Google) provide conversational tutoring at scale but operate statelessly within sessions, each interaction begins without knowledge of the learner's history, preferences, or prior struggles. Pardos and Bhandari [22] examined learning gain differences with ChatGPT-based tutoring, noting the stateless nature as a structural limitation. AgentTutor [17] demonstrated multi-turn personalised teaching with dynamic strategy adjustment using curriculum decomposition and knowledge memory modules. Sudar's AI tutor addresses this gap through longitudinal cross-session memory: every tutor exchange is logged and summarised in the learner's *ai_tutor_context*, so the tutor always knows who the learner is, what they have struggled with, and how they prefer to be taught.

2.5 AI-Augmented Textbooks

The LearnLM team's Learn Your Way [16] transforms source material with personalisation (grade level, interests), multiple views (immersive text, slides, audio, mind maps), and formative assessment; a randomised trial showed gains over a digital reader. However, Learn Your Way focuses on augmenting a single textbook within one product: it does not claim a longitudinal learner model across sessions and courses, nor does it provide an architecture to augment existing LMSs, the gap ALP is designed to fill.

2.6 xAPI, LRS, and the Interoperability Gap

The Experience API (xAPI) specification [1] standardises how learning experiences are recorded and reported across systems. xAPI and its Learning Record Store (LRS) ecosystem standardise event reporting but provide no intelligence layer, no learner model, no adaptive sequencing, and no tutoring. ALP provides the missing intelligence layer on top of any xAPI-capable LMS.

2.7 Comparative Analysis

Table 1 contrasts Sudar against representative systems across key capability dimensions. The cost column uses empirically observed Q1 2026 pricing at 1,000 learners per month (see Section 5 for methodology).

Feature	Sudar + ALP	Agent Tutor	Khan migo	Learn Your Way	Typical LMS+AI	Cost (1k/mo)
Learner model	Longitudinal Twin	Session-scoped	Stateless	Per-session	None	
Tutor memory	Cross-session, cross-course	Within session	Stateless	Not claimed	Stateless	
Open learner model	✓	–	–	–	–	
Modalities	6 (read/listen/video/ podcast/map/cards)	Text	Text	Text/slides/audio/map	Text/video	
Augments existing LMS	✓ (ALP)	–	–	–	N/A	
Open source	✓ Apache 2.0	–	–	–	Rarely	
Cost (1,000/mo)	~$21	N/A	~$4,000	N/A	$3,400–$44,000	

Table 1. Capability comparison of Sudar + ALP with representative systems. Khanmigo individual ~$4/learner/month [35]. Cost column: AI infrastructure only at 1,000 learners/month, Q1 2026 (see Section 5 for full methodology).

3. System Design

3.1 Platform Architecture

The Sudar platform has three surfaces sharing one data layer: **Studio** (admin/creator), **Learn** (learner interface), and **Intelligence** (backend engine). All three share a single source of truth, a Supabase/PostgreSQL instance, for authentication, learner profiles, content, and events. Studio and Learn are Next.js 15 applications (TypeScript strict mode, App Router, Tailwind CSS); Intelligence is a Python 3.11+ FastAPI service. The AI provider layer is abstracted behind an environment-variable-configurable fallback chain (Together AI → OpenRouter → OpenAI → Anthropic), enabling substitution without code changes.

Shared Data Layer - Supabase / PostgreSQL + pgvector | Auth | Learner Profiles | Content | Events

Studio (Admin / Creator) Learn (Learner Interface) Intelligence (Backend Engine)

Course authoring Personalised dashboard Adaptive engine

AI content generation 6-modality course viewer Tutor engine

Analytics & compliance AI tutor sidebar Modality dispatcher

Studio governance Progress & certifications TTS / event processing

■■■ solid = write/query ■■■ ■■■ dashed = async profile update ■■■

Figure 1. Sudar reference platform - three surfaces over a shared Supabase/PostgreSQL data layer. Solid arrows show synchronous write/query flow; dashed arrows show asynchronous profile updates from the Intelligence engine.

3.2 Digital Learner Twin

A persistent **Digital Learner Twin (DLT)** is stored in the *learner_profiles* table. The DLT maintains a unified profile accumulating behavioural, preference, and contextual signals across all sessions and courses. It distinguishes between *observed signals* (quiz scores, time-on-task, modality selections) and *inferred signals* (modality affinity from completion patterns, difficulty comfort from quiz trajectories). A newly created profile is initialised with neutral affinity (0.5 across all formats); an onboarding preference capture seeds it before adaptive behaviour begins.

The learner can inspect and correct their own model at any time via the '*Your context*' panel, a live, editable view supporting GDPR Articles 15–16 and FERPA compliance (see Section 3.9). Modality affinity scores are estimates from behavioural proxies; a learner who selects a suboptimal modality will accumulate affinity scores reflecting choice rather than outcome, a known limitation addressed in the pilot study plan.

Observed Signals Digital Learner Twin Outputs

Quiz scores ■■■■■■ Learning style & modality affinity ■■■■■■ Next-best-action

Time-on-task Engagement & behavioural signals Tutor style

Modality choice AI tutor context & interaction log Sequence order

Preferences Cognitive style & difficulty comfort Adaptive nudges

Cold start: neutral 0.5 affinity → onboarding capture → data-driven personalisation

Figure 2. Digital Learner Twin data model - observed inputs, inferred signals, and adaptive outputs. The Twin persists across all sessions and courses.

3.3 AI Tutor with Longitudinal Memory

The AI tutor Sudar is both *reactive* (retrieval-augmented generation over course content to answer learner questions) and *proactive* (generating nudges when struggle or prolonged pause is inferred from behavioural signals). Every exchange is logged in *ai_interactions*. The tutor's system prompt includes the learner's memory summary drawn from *ai_tutor_context*, so responses reference prior struggles, match the learner's preferred explanation style, and explicitly connect new material to prior learning. Most AI in LMS implementations are stateless within a session [29]; Sudar maintains memory across sessions and courses.

Key tutor interface features:

- **Text-selection popup** - highlight any passage to invoke six pre-built actions (*Explain this, Give me an example, Simplify this, Summarise, Why does this matter?, How does this connect…*) or a custom question.
- **Resizable overlay panel and global floating chat** - accessible on any page, not only course pages.
- **Proactive tap-to-reply nudges** - surfaced on the dashboard and course viewer with optional reply chips that forward to the same tutor query pipeline.
- **Generative block renderer** - tutor can respond with structured artefacts: quiz questions, course recommendation cards, multi-step workflow trackers, and action groups.
- **"My Memory" page** - insights carousel showing what Sudar has learned about the learner, supporting metacognitive reflection.

3.4 Modality-Agnostic Delivery

Content is authored once and delivered in six fully implemented modalities, selectable per learner in real time. A modality dispatcher in the Intelligence layer recommends the next modality from the learner's affinity scores, operationalising dual-coding principles [18, 8]. Which modalities are active depends on course configuration: video scenes and podcast dialogue are optional authored fields; SCORM packages bypass the tab switcher entirely.

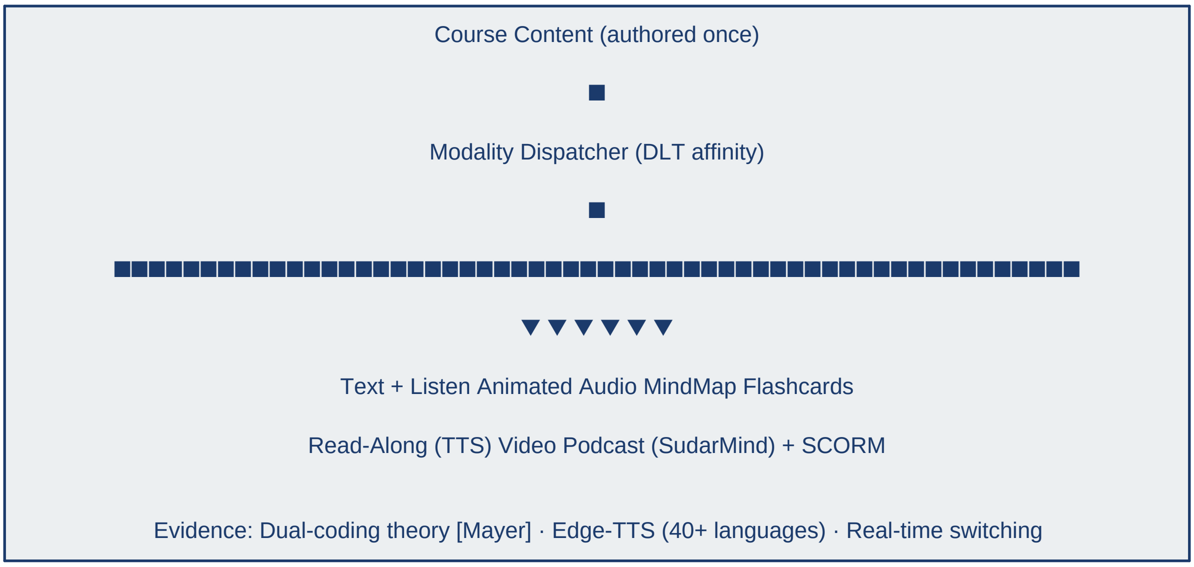

Figure 3. Modality-agnostic delivery stack - content authored once, dispatched across six formats by DLT affinity scores.

The six modalities are:

- **Text with Read-Along** - Sentence-level TTS read-along via Edge-TTS (40+ languages, 300+ voices, zero infrastructure cost). Each sentence is highlighted in place as it plays.
- **Listen (standalone audio TTS)** - Dedicated audio tab for on-demand full-module TTS with optional regeneration and learner voice preferences.

- **Animated Video** - Scene-by-scene animation rendered entirely in-browser with four styles: kinetic, word-reveal, fade, list. Richer video can be produced via the SudarVid microservice when configured.
- **Audio Podcast** - Two-voice dialogue (host + domain expert) with live transcript and jump-to-segment playback.
- **MindMap (SudarMind)** - Interactive concept map via POST */api/ai/generate-mindmap*. Eight colour families, Bézier curves, module or course scope.
- **Flashcards + SCORM** - On-demand review cards supporting spaced repetition [23, 5]. SCORM 1.2 packages render in sandboxed iframe; scores write to *learning_events*.

3.5 Adaptive Sequencing and Quiz Architecture

Learning paths support mandatory and optional courses with unlock rules. For optional courses, ordering is adapted by the Intelligence engine. The **next-best-action** algorithm: (1) filters to enrolled, incomplete items; (2) prioritises mandatory items and approaching due dates; (3) ranks optional items by cosine similarity of skill tags to knowledge gaps multiplied by modality affinity; (4) surfaces the top-ranked item with a human-readable rationale.

Five quiz mode archetypes reduce habituation and vary retrieval modality [23]:

- **Standard closed-response** - conventional MCQ or short-answer.
- **Predict-then-learn** - learner commits to a prediction before seeing content; activates prior knowledge and increases encoding depth [5].
- **Confidence-tagged** - learner self-reports certainty alongside each answer; supports metacognitive calibration [33].
- **Scenario-fork** - branching case study where learner choices determine the scenario path; models real-world decision-making.
- **Peer-contrast** - presents anonymised responses from other learners before the learner finalises their own answer.

3.6 Agent Orchestration - SudarAgents

Beyond conversational tutoring, Sudar exposes a gateway for bounded plans with a restricted tool catalogue. Reads and scoped writes reuse the Twin, *learning_events*, and rollups. Each completed mission records structured output plus tool traces in *agent_runs*, providing an auditable trail. Representative missions include learner week-plan sketches and cohort path diagnostics. Org policy can disable individual capabilities; the HTTP contract is documented in *docs/AGENTS_PLATFORM.md* and exposed at */api/agents/skills* for partner backends.

This design operationalises recommendation and light reflection loops aligned with formative practice and calibration principles [5, 33], without claims of new empirical efficacy relative to unstructured chat.

3.7 Plugin Architecture - The Adaptive Learning Layer (ALP)

The ALP is the paper's most novel architectural contribution: a plugin layer packaging Sudar's capabilities as independently deployable services for existing LMSs. Organisations running Moodle, Canvas, Blackboard, or a proprietary LMS can gain learner memory and adaptive tutoring without abandoning their infrastructure investment. Moodle 4.5 introduced a dedicated AI subsystem [20]; ALP is architecturally aligned with this model. Moodle alone has over 500 million registered users across 233 countries [21].

Host LMS (Moodle / Canvas / Blackboard / Sakai)

Content | Enrolments | Gradebook (LTI AGS)

■ xAPI events / webhooks

▼

■■■■■■■■■■■■■■ ALP - Sudar Intelligence ■■■■■■■■■■■■■■

■ SudarMemory SudarChat SudarRecommend ■

■ SudarStudio Embed SudarAdapt ■

■ SudarAgents ■

■■■

■ Next-best-action · tutor response · modality recs

▼

Sudar DB - Digital Learner Twins

Transport: HTTPS + per-institution API keys | LTI 1.3 Advantage | PII-free pseudonymised events

Figure 4. ALP integration architecture. The host LMS sends learning events to SudarMemory; ALP returns next-best-action, tutor responses, and modality recommendations. The LMS retains ownership of content and enrolments.

Five independently deployable ALP plugins (reference vs. roadmap status noted):

- **SudarMemory** *(reference)* - Foundational layer. Intercepts xAPI or webhook events; maintains a Digital Learner Twin per user via */api/alp/events*. Zero disruption to existing LMS workflows.
- **SudarChat** *(reference)* - Embeds the Sudar AI tutor via */api/alp/tutor/query* as a sidebar or modal inside any LMS course page. Every message queries the learner's Twin from SudarMemory.
- **SudarRecommend** *(reference)* - Next-best-action widget for the LMS dashboard via */api/alp/next-action*; no schema changes required on the host database.
- **SudarStudio Embed** *(roadmap)* - Adds 'Generate with AI' to the host LMS course editor for mindmaps, flashcards, audio narration, and animated video storyboards.
- **SudarAdapt** *(roadmap)* - Integrates with the LMS's conditional activity system to dynamically unlock or reorder optional content. Sudar Learn already supports adaptive path ordering for Sudar-hosted paths.

Security and data isolation: ALP communicates over HTTPS using per-institution API keys. No host LMS student PII is transmitted beyond pseudonymised learning events (user ID hash, event type, timestamp, payload). For Moodle and Canvas, ALP leverages LTI 1.3 Advantage [34] for secure tool embedding, token exchange, and scoped data access. Packaging the same contract as distributable installable modules per LMS vendor, and hardening for multi-tenant production, remain deployment engineering tasks documented in *docs/ALP_API.md*.

3.8 Open-Source Extensibility

Sudar's modality delivery stack is designed for community extension and backend substitution. The *VideoScene[]*, *DialogueSegment[]*, and *ModalityVariants* types are provider-agnostic interfaces; any modality backend can be swapped without touching the delivery layer. Video backend options range from the default zero-cost in-browser renderer to Higgsfield [13] (cinema-grade multi-model platform), Wan 2.1 [31] (MIT open-source, 1.3B–14B parameters), and HunyuanVideo (Tencent, Diffusion Transformer). A school in Sub-Saharan Africa can self-host Wan 2.1 on a local GPU at zero per-call cost while the learner experience, Digital Learner Twin, adaptive sequencing, AI tutor, remains identical.

The Apache 2.0 licence enables: new modality plugins (VR, AR, 3D simulation); new LLM backends via any OpenAI-compatible API endpoint; new language and voice packs (Sarvam AI Indian English neural voices are already integrated); and vertical-specific forks for medical education, software engineering bootcamps, and regulated-industry compliance.

3.9 Intelligence Layer Evolution

The current Intelligence layer implements reactive Q&A via RAG, rule-based proactive nudges, and a longitudinal memory updated from quiz outcomes and tutor exchanges. As the *learning_events* table accumulates data, the engine is designed to transition from rule-based heuristics to learned intervention policies. Research demonstrates RL-driven ITS achieves 28.6% better intervention adaptability, 31.2% reduction in recurring errors, and 24.8% lower dropout rates versus rule-based systems [19].

Phase 1 - Current	Phase 2 - Planned	Phase 3 - Future
RAG Q&A; tutor Rule-based nudges SudarAgents (bounded) Longitudinal memory	RL policy learning Predictive interventions Learned modality recs Pilot study	SudarCoach SudarAssess (IRT) SudarSocial / Offline Community plugins

Figure 5. Intelligence layer evolution - from rule-based heuristics (current) to RL-driven policy learning (planned) and community intelligence plugins (future).

3.10 Privacy Architecture and Data Governance

The Digital Learner Twin accumulates behavioural and contextual data across sessions. Sudar's privacy architecture rests on three principles:

- **Data minimisation** - Only behavioural signals directly relevant to learning adaptation are stored: modality affinity, quiz outcomes, time-on-task, and tutor interaction context. Biometric, financial, and demographic data are not collected.
- **User inspection and correction** - The "Your context" panel exposes the full learner model at any time, supporting GDPR Articles 15–16 and FERPA rights.
- **Right to erasure** - The Digital Learner Twin can be reset or deleted at any time (GDPR Article 17 / EU AI Act [11]). Deletion cascades across *learner_profiles*, *ai_interactions*, and *learning_events*.

4. Implementation and Reproducibility

The reference platform is implemented with Next.js 15 (Studio, Learn; TypeScript strict mode, App Router, Tailwind CSS), Python 3.11+ FastAPI (Intelligence; async handlers, Pydantic v2), and Supabase (PostgreSQL + pgvector, auth, storage). AI providers include Together AI (primary; open-weight models), OpenAI, and Anthropic via a configurable fallback chain. TTS uses Edge-TTS (default; zero cost, 40+ languages, 300+ voices) with optional Sarvam AI Indian English neural voices. A live demonstration is accessible at *[demo link redacted for review]*. Full schema, API surface, and event model are documented in ECOSYSTEM.md and RESEARCH_FOUNDATION.md in the repository [14].

Currently implemented features include: shared authentication and schema; course and learning path CRUD with publish workflow; document-to-course generation and SCORM 1.2 import; learner dashboard with streak, time-on-task, Sudar recommendations, and upcoming deadlines; multimodal course viewer (all six modalities per Section 3.4); AI tutor with longitudinal memory, RAG, text-selection popup, resizable overlay, tap-to-reply proactive nudges, "Your context" live memory panel, and generative block rendering; five quiz mode archetypes and five lesson archetypes; next-best-action and adaptive path ordering; learning paths with unlock rules; shareable public certificate pages with browser print/save and server-generated PDF download; optional per-enrollment generative overlays with org policy and consent; Studio governance and technical trust pack; compliance view (overdue/at-risk/on-track) and email reminders; analytics rollups and risk signals; global search; and learner preferences. ALP reference endpoints are implemented on the Learn application (Section 3.7). Production deployment is documented: Studio and Learn on Vercel; Intelligence on Railway, Render, or Fly.io.

Table	Key Fields	Purpose
learner_profiles	ai_tutor_context (JSON), next_best_action (JSON), modality_affinity (JSON, 0.0–1.0), streak_days, avg_session_duration, preferences (JSONB)	Digital Learner Twin
learning_events	event_type (module_start/complete, quiz_attempt, video_play, section_heartbeat, ai_tutor_query, modality_switch, drop_off), payload, modality, duration	Behavioural signal log
ai_interactions	user_message, ai_response, context_used, helpful (bool)	Tutor exchange history
enrollments	path/course, progress_pct, due_date, status, personalization_overlays (JSONB)	Enrolment tracking
certifications	issued_at, expires_at, verification_code	Shareable cert + PDF
skills / learner_skills	skill tags, gap scores	Competency knowledge graph
agent_runs	mission_type, tool_traces (JSON), structured_output (JSON), created_at	SudarAgents audit trail

Table 2. Key schema tables for reproducibility. Full schema and migration files available in supabase/migrations/ in the repository [14].

5. Economic Accessibility: The Radical Cost Case

The economic dimension of Sudar is not incidental to its design, it is one of its three primary contributions. Operating Sudar at full engagement, 10 courses generated, 20 learner review sessions with AI tutor chat, adaptive personalisation, mindmap and flashcard generation, and TTS narration, costs approximately **$0.20 per week** on the Together AI + Edge-TTS stack. All figures reflect pricing as observed in Q1 2026; the cost floor continues to fall with Llama 4 and equivalent 2026-era models [4].

5.1 Infrastructure Cost Comparison

Stack	Per Learner / Month	Per Month (1,000)	Annual (1,000)
Sudar - Together AI 8B + Edge-TTS	$0.021	$21	$252
Sudar - Together AI 70B + Edge-TTS	$0.105	$105	$1,260
Self-hosted (Ollama + open-weight, GPU)	~$0*	~$0*	~$0*
GPT-4o + OpenAI TTS-1	$3.41	$3,405	$40,860
Claude 3.5 Sonnet + Azure TTS	$3.74	$3,735	$44,820
Docebo (platform licence, low end)	$5.83+	$5,830+	$69,960+
Sana Labs (est., min. 300 seats)	~$15	~$15,000	~$180,000

*Table 3. Infrastructure cost comparison per 1,000 learners per month (Q1 2026). *Self-hosted cost ~$0 per API call after hardware provisioning; excludes amortised GPU hardware. AI infrastructure only; hosting excluded. Supabase free tier: $0; Pro: $25/month. Studio/Learn on Vercel free tier: $0. Intelligence on Railway/Render: ~$5–$10/month at moderate traffic.*

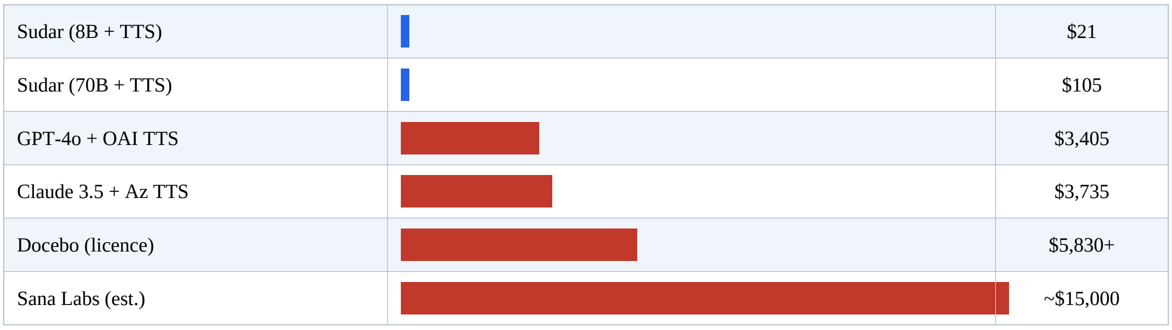

Figure 6. Infrastructure cost per 1,000 learners per month (Q1 2026). Blue bars indicate Sudar open-weight stacks; red bars indicate proprietary/commercial stacks. Sudar's open-weight stack achieves >99% cost reduction.

5.2 The Inference Cost Collapse

AI inference costs have fallen approximately 1,000-fold from 2022 to 2026 at roughly 10× per year [4], driven by GPU efficiency gains, software optimisation (vLLM, TensorRT-LLM), Mixture-of-Experts architectures, and quantisation. GPT-3-class performance costs $0.06 per million tokens today versus $60 in 2021. The arrival of open-weight frontier models (Llama 4 and equivalents in 2026) extends this trajectory further. The cost of delivering a world-class personalised tutor interaction to one learner is now less than $0.02 per month, below the cost of a single SMS message. The economic argument against AI-native learning at scale has expired.

5.3 Education Access and Sudar's Response

As of 2024, 251 million children and youth remain out of school globally [28]. 2.6 billion people remain offline [37], with mobile internet in Africa costing 14× more than in Europe. An annual financing gap of $39 billion exists for basic education in low- and lower-middle-income countries through 2030 [28].

Sudar's marginal AI infrastructure cost of $0.21/user/month ($2.52/year) is reported for comparison with other API stacks, it is not directly comparable to all-in training spend benchmarks [$1,283/employee/year; 36]. High-end AI learning platforms priced at $12–$20 per user per month are not deployable in resource-constrained contexts. A platform whose full AI infrastructure costs $0.02 per learner per month, and whose stack is open-source, self-hostable, and runnable on open-weight models, is. For connectivity-constrained scenarios, the relevant deployment path is the offline-first self-hosted variant (Wan 2.1 or a quantised open-weight LLM on a local GPU server), a planned capability (Section 7.2) enabled by Sudar's modality backend abstraction.

6. Evidence and Evaluation Strategy

6.1 Research Foundation

The design is evidence-informed. RESEARCH_FOUNDATION.md in the repository [14] maps each capability to learning-science references: adaptive instruction [28, 3], the two-sigma problem [6], dual coding [18, 8], self-regulated learning [33], formative assessment [5, 23], intelligent tutoring [28, 22], open learner models [7, 15], and learner modelling [22, 7].

6.2 Scope of Current Claims

This is a system and architecture paper. Current claims are: (a) the platform implements all described components at the level of a working, deployed proof of concept, accessible at *[demo link redacted for review]*, with explicit caveats for course-dependent modalities, optional product layers, and ongoing LMS packaging; (b) each component is grounded in established learning-science evidence; (c) the ALP architecture addresses a gap no widely adopted technology currently fills [29]; and (d) the economic cost structure has been empirically observed and validated against current provider pricing as of Q1 2026.

6.3 Planned Pilot Study

A pilot evaluation is planned following the current build phase, conducted in partnership with one or more institutional partners (currently under negotiation; outcomes to be reported in a subsequent paper). Research questions:

- Does Sudar's longitudinal tutor memory improve learner satisfaction and self-reported learning experience relative to a stateless tutor baseline?
- Does adaptive sequencing (next-best-action) reduce time-to-competency compared to fixed-order course delivery?
- Does ALP plugin integration produce measurable engagement improvement (completion rates, session frequency, tutor interaction depth) on an existing Moodle or Canvas installation?

Design: Mixed-methods. Quantitative: pre/post knowledge assessments, module completion rates, time-on-task, quiz score trajectories. Qualitative: semi-structured interviews. Participants: 50–200 learners, one partner institution, 4–8 weeks, ≥5 courses per learner. IRB/ethics board approval will be obtained prior to data collection. Participants retain the right to withdraw at any time and to request deletion of their Digital Learner Twin. Protocol details are documented in *docs/research/PILOT_PROTOCOL.md*.

7. Discussion

7.1 Limitations

- **Absence of empirical efficacy data.** Controlled efficacy data are not yet available; claims about learning gains remain grounded in the underlying learning-science evidence rather than Sudar-specific trials. The pilot study described in Section 6.3 is planned to address this directly.
- **ALP connector implementation status.** The ALP API surface is fully defined and documented in ECOSYSTEM.md; packaging as one-click LMS-marketplace add-ons for Moodle, Canvas, and Blackboard, and validating identity mapping in diverse SSO environments, remains deployment engineering work.
- **Digital Learner Twin accuracy.** The DLT relies on inferred signals that may not fully capture a learner's actual knowledge state [22, 7]. Modality affinity scores are estimates from behavioural proxies; a learner who selects a suboptimal modality will accumulate affinity scores reflecting choice rather than outcome.
- **Edge-TTS licensing.** Edge-TTS accesses Microsoft's Azure Neural Voice infrastructure without a formal commercial licence; production deployments with paying customers should provision a licensed TTS service (Azure AI Speech, Sarvam AI, or equivalent).
- **Data portability.** The ability for a learner to export their Digital Learner Twin and import it into another platform is not yet implemented; a formal data export and import API is in the roadmap.
- **Learner twin interpretability.** Educators and administrators may not understand or trust AI-generated inferences about students. Roumpas et al. [24] provide an ethical framework for

cognitive-aware human-AI interaction in multimodal adaptive systems; full alignment, including formal auditability of AI-generated inferences, remains future work.

- **Predictive capabilities require longitudinal data.** The Intelligence layer today combines rules with heuristics; RL-style policies require sufficient event data to learn reliably. Phase 2 (Section 3.9) explicitly depends on accumulated learning event streams.

7.2 Future Work

Planned work includes: LMS-specific connector implementations for ALP (Moodle 4.5+, Canvas, Blackboard); completing SudarPlay (gamified modality) and SudarFeed (social feed); developing the predictive intervention layer with RL-based policy learning; running controlled efficacy studies per *docs/research/PILOT_PROTOCOL.md*; releasing SudarCoach, SudarAssess (IRT psychometric layer), and SudarCompliance as first-party Intelligence plugins; building the community plugin ecosystem; supporting enterprise features (SSO, HRIS integration); developing an offline-first variant for connectivity-constrained environments; and implementing the Digital Learner Twin data portability API (cross-platform export/import).

SudarSocial is in early development as an integration with the open-source virtual world platform WorkAdventure, enabling peer learning orchestration within persistent collaborative learning environments. The Digital Learner Twin will inform peer-matching algorithms, supported by socioconstructivist learning theory [30].

8. Conclusion

We presented Sudar, an AI-native learning platform making three contributions: (1) a reference platform (LAMP) combining authoring, delivery, and intelligence around a Digital Learner Twin, adaptive sequencing, a complete multimodal delivery stack, bounded agent orchestration, and an AI tutor with longitudinal cross-session memory, open source, deployed, and accessible at [demo link redacted for review]; (2) the Adaptive Learning Layer (ALP), a plugin architecture with the potential to reach the hundreds of millions of Moodle, Canvas, Blackboard, and Sakai users [21] without requiring LMS replacement; and (3) an empirically observed economic model reducing the infrastructure cost of AI-native personalised learning to less than $0.02 per learner per month, viable at the scale of national education systems.

The inference cost collapse [4] is the macro condition that makes all of this possible. Bloom's two-sigma problem [6] has been known since 1984; the economic barrier to solving it at population scale has only recently expired. The paper is honest about what Sudar does not yet have: controlled efficacy data, production LMS-specific connectors, and a completed offline-first deployment path. These are the next milestones, not reasons to delay the architectural and economic contributions presented here.

Call for the Community - The reference implementation is open source (Apache 2.0), grounded in a broad learning-science evidence base, and designed as a bedrock upon which the community can build modalities, intelligence plugins, and LMS connectors. Institutions, researchers, LMS administrators, and open-source contributors are invited to engage: use Sudar in your courses, build ALP connectors for your LMS, extend the modality stack, and contribute to the pilot study programme. The goal is a world where adaptive, memory-aware learning is not a privilege of well-funded institutions but open infrastructure, like the web itself.

Acknowledgments

Sudar is the author's independent project, designed and implemented as a solo effort across the Studio, Learn, and Intelligence surfaces. The system and research are available at [repository link redacted for review]. Institutions, organisations, and open-source contributors are invited to collaborate on pilots, plugin integrations, and community extensions. Contact: [contact redacted for review]

Appendix A - Schema Summary

Key tables for reproducibility. Full schema, migration files, and event model: ECOSYSTEM.md and *supabase/migrations/* in the repository [14].

Table	Key Fields	Purpose
learner_profiles	ai_tutor_context (JSON), next_best_action (JSON), modality_affinity (JSON, 0–1), streak_days, avg_session_duration, preferences (JSONB)	Digital Learner Twin
learning_events	event_type, payload, modality, duration	Behavioural signal log
ai_interactions	user_message, ai_response, context_used, helpful (bool)	Tutor history
enrollments	path/course, progress_pct, due_date, status, personalization_overlays (JSONB)	Enrolment tracking
certifications	issued_at, expires_at, verification_code	PDF cert links
skills / learner_skills / skill_gaps	skill tags, gap scores	Competency knowledge graph
learner_performance_records	institution-aware KPI and grade tracking	Analytics / compliance
agent_runs	mission_type, tool_traces (JSON), structured_output (JSON), created_at	SudarAgents audit trail

Appendix B - Infrastructure Cost Reference

All figures observed Q1 2026. AI provider pricing changes rapidly; verify current rates at time of deployment. AI infrastructure only; hosting costs excluded.

Stack	Per Learner / Month	Annual (1k)
Together AI 8B + Edge-TTS	$0.021	$252
Together AI 70B + Edge-TTS	$0.105	$1,260
Self-hosted (Ollama + open-weight, GPU)	~$0*	~$0*
GPT-4o + OpenAI TTS-1	$3.41	$40,860
Claude 3.5 Sonnet + Azure TTS	$3.74	$44,820
Docebo (platform licence, low end)	$5.83+	$69,960+
Sana Labs (est., min. 300 seats)	~$15	~$180,000

*Self-hosted ~$0 per API call after hardware provisioning; excludes amortised GPU hardware. Supabase free tier: $0; Pro: $25/month.

Appendix C - Reproducibility Artefacts

The following documents are available in the repository [14]:

- **docs/ALP_API.md** - ALP HTTP endpoint reference, field mappings, and integration examples.

- **docs/AGENTS_PLATFORM.md** - SudarAgents mission catalogue, tool catalogue, and security model.
- **docs/research/COST_WORKSHEET.md** - Versioned cost assumptions; fill and update at time of deployment.
- **docs/research/EVALUATION_APPENDIX.md** - Scope and limitations for each empirical claim.
- **docs/research/PILOT_PROTOCOL.md** - IRB-ready pilot study protocol.
- **scripts/benchmark-sudar.mjs** - Automated latency and token measurement script.
- **ECOSYSTEM.md** - Full schema, event model, and API surface.
- **RESEARCH_FOUNDATION.md** - Evidence-to-feature mapping.

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